

Reflection

AI Career Compass | What the data supports and cannot prove

Snapshot	2026-09-21
Postings / companies	852 / 107
Source checks	198

What this version supports

The snapshot supports descriptive comparisons within the observed postings. For example, 304 of 852 postings name at least one location outside California, and 42 explicitly identify a graduate pathway. These counts help students and university career groups compare places and separate graduate evidence from broad seniority labels. They do not measure the number of open hires or the total US AI market.

A design change prompted by the data

The earlier any-skill measure counted 682 of 852 postings after selecting Python and Machine learning. That gave too much credit to broad labels. The current version requires at least 70% of each posting's distinct tracked skills. The same two-skill profile covers 74 of 816 scored postings; adding LLMs raises it to 167, while another 247 move closer. The separate crossing and progress counts make the learning experiment more informative.

What the target test establishes

Synthetic profiles and three-step learning paths across five directions compared targets from 40% to 80%. Fixed 70% asks for broader coverage than 60% while preserving incremental gains. It is a transparent product choice, not an empirically validated hiring cutoff. Profile composition, short skill lists and equal weighting affect the results. No student outcome study or employer validation was conducted.

What the data cannot prove

The 795 automated-discovery records are not equivalent to full source review. The 57 curated records also cannot establish current recruiting intent. The selected boards overrepresent reachable technology employers and inherit geographic and topic bias. Unknown experience or skills do not mean zero requirements. A mention may be optional or an alternative, and confidence labels are not probabilities. Coverage cannot establish proficiency, eligibility, interview chances, salary expectations or the causal benefit of a course.

Recommendations and next evidence

Career groups can include non-California locations in exploration sessions because 304 sampled postings name one. They can also run role-specific learning sessions: SQL appears in 66 of 79 Data Science postings, while Kubernetes appears in 66 of 183 ML Infrastructure postings. A useful next study would independently review a stratified sample of extracted skills and ambiguous experience labels, then test whether students understand the distinctions. That evaluation remains future work.

AI assistance and data handling

AI agents assisted with collection, normalization, interface implementation and testing. Source IDs, URLs, review labels and saved corrections make the claims inspectable. The interface collects no sensitive applicant data, and skill selection remains local to the browser.

Sources

[Official posting example \(checked 2026-09-21\)](#)

[Source URLs and collection dates](#)

[Code, collection scripts and frozen source partitions](#)